Automotive BLE-to-CAN Hardware Gateway

KICAD 10 · 4-LAYER

High-Speed CAN FD Telematics Module with Seamless Li-ion Battery Failover & Wide-VIN Protection

Designer: **Ajith Shajan** (Model Engineering College)

Bus: **CAN FD (5 Mbps)**

Radio: **BLE 5.3 (2.4 GHz)**

1. VEHICLE POWER CONDITIONING

9V – 36V DC

Automotive Input Stage

ISO 7637-2

- **2A Fuse (F1)** + **36V 400W TVS (D1)** load-dump clamp
- **-60V P-MOSFET (Q1)** reverse polarity clamp (<15 mW dissipation)
- Rail: `VIN_PROTECTED` (9V–36V transient protected)

2. POWER TREE & BATTERY BACKUP

Auto Failover

TI LM5164 Wide-VIN Buck

100V → 3.3V (1A)

Primary vehicle supply generating `3V3_REG` logic rail.

TI BQ24040 Li-Ion Charger

5V → 4.2V (250mA)

Fed by dedicated 65V buck (U4). Charges 3.7V Li-ion battery with NTC thermistor thermal foldback.

TI TPS63900 Buck-Boost

3.0V–4.2V → 3.3V

High-efficiency backup regulator generating `3V3_BAT` from single-cell battery.

TI TPS2116 Fast Power Mux

< 2 µs Failover

Hardwired priority on vehicle power (`3V3_REG`). Automatically switches to battery backup (`3V3_BAT`) seamlessly upon engine cranking or vehicle power cut.

3. WIRELESS PROCESSING & RF FRONT-END

TI CC2340R52

Arm® Cortex®-M0+ Core

48 MHz

- 512 KB Flash, 36 KB SRAM, Ultra-low-power standby mode
- Dual high-stability crystals: **48 MHz** main + **32.768 kHz** RTC
- SWD programming & debugging interface (J3)

Bluetooth Low Energy 5.3 Radio

2.4 GHz (+8 dBm)

Streams real-time vehicle CAN telemetry, diagnostic alerts, and sensor logs to mobile devices and edge telematics units.

50Ω RF Matching & Antenna

CPWG + Chip Ant

- **Coplanar Waveguide with Ground (CPWG):** 50Ω controlled impedance over unbroken Layer 2 GND
- **Discrete Pi Filter:** C19, L3, C21, L4 for harmonic attenuation
- **Infineon ESD112:** Low capacitance (<0.3 pF) RF ESD clamp
- **Johanson 2450AT:** Omnidirectional ceramic chip antenna

⟵ High-Speed SPI (10 MHz) + nINT + WAKE ⟶

4. HIGH-SPEED CAN FD SUBSYSTEM

TCAN4550-Q1

TI TCAN4550-Q1 Controller + Transceiver

ISO 11898-2:2016

- Integrated Bosch CAN FD controller + physical layer transceiver
- High-speed data rates up to **5 Mbps**
- Automotive grade **AEC-Q100 Grade 1** (-40°C to +125°C)
- 40 MHz clock reference for high-resolution timing

Dedicated CAN Power Buck (U8)

TI LMR36506 (6V)

Supplies isolated ~6.0V `VSUP_CAN` directly to the transceiver. Isolates sharp CAN switching currents from the master 3.3V logic supply.

Bus Hardening & EMC Protection

CISPR 25 Class 5

- **Common-Mode Choke:** Coilcraft ACT45B (51 µH)
- **Split Termination (JP1):** Dual 60.4Ω + 4.7nF center-tap capacitor for RF noise absorption (Jumper selectable 120Ω)
- **Automotive ESD:** Nexperia ESDCAN24-2BWY (±30kV contact/air)

Vehicle CAN Bus Terminal (J2)

Screw Terminal

Connects to vehicle OBD-II / J1939 network (CAN_H, CAN_L, Chassis GND).

Stackup: 4-Layer FR4 (~1.6mm) · JLC04161H (1oz / 0.5oz Cu) • **Verification:** DRC Clean (0 Errors) · ERC Clean (0 Warnings) • **RF Impedance:** 50Ω CPWG ±5% • **Operating Temp:** -40°C to +85°C (AEC-Q100/101/200)